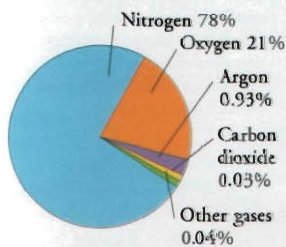


ATMOSPHERE



LIFE ON EARTH could not exist without Earth's atmosphere. The atmosphere is a colourless, tasteless, odourless blanket of gases that surrounds the Earth. It gives us air to breathe and water to drink. As well as keeping us warm by retaining the Sun's heat, it also shields us from the Sun's harmful rays. The atmosphere is approximately 700 km (440 miles) deep, but it has no distinct boundary. As it extends into space, it becomes thinner, eventually fading out. Human activity is upsetting the atmosphere's natural balance, with damaging results.



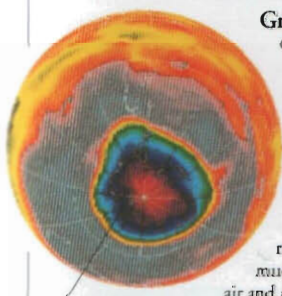
Pie chart showing the composition of the atmosphere.

Composition of the atmosphere

Earth's atmosphere is made mainly of two gases – nitrogen and oxygen. It also contains small amounts of argon and carbon dioxide, with tiny traces of other gases. The oxygen is made primarily by green plants, which maintain the balance of gases.

Ozone layer

The thin layer of ozone gas within the stratosphere protects us by absorbing harmful ultraviolet rays from the Sun. But build-up of man-made gases called chlorofluorocarbons (CFCs) has depleted the ozone layer, and holes have started to appear in it every spring over the poles.



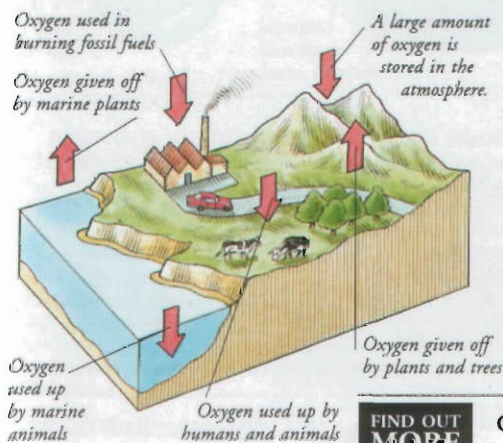
Greenhouse effect

Carbon dioxide and other gases in the atmosphere act like glass in a greenhouse, trapping the Sun's heat. This "greenhouse effect" keeps the Earth warm. But human activity, such as burning forests and running cars, releases too much carbon dioxide into the air and may cause global warming.

Ozone hole over Antarctica is shown as violet and pink



Some aerosol sprays use CFC gases.



Oxygen cycle

Gases continually circulate between the atmosphere and living things. Animals breathe in oxygen to help them release energy from food, and breathe out carbon dioxide. Green plants release oxygen back into the air and take in carbon dioxide as they absorb energy from the Sun. Oxygen is also used in the burning of fossil fuels.

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Layers of the atmosphere

The atmosphere is divided into five different layers. The composition of gases varies within these layers, as does the temperature which drops in the troposphere, the lowest layer, and rises in the stratosphere above.



James Glaisher

English meteorologist James Glaisher (1809–1903) was one of the many balloonists who, during the 19th century, took great risks when they ascended to extraordinary heights to discover more about the atmosphere. Glaisher went up almost 12 km (7.5 miles) into the troposphere without oxygen or protective clothing. Such research led to the discovery that air becomes cooler with altitude.

